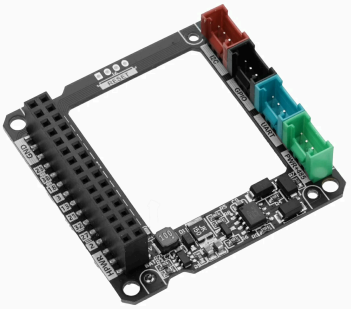
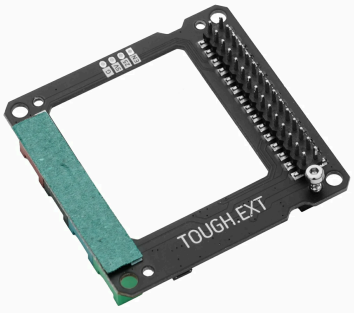
ToughC5

SKU:K162





Description

ToughC5 is an outdoor waterproof IoT controller. Built around the ESP32-C5HR8 core module, it features a **RISC-V** 32-bit single-core processor running at 240MHz, with 16MB Flash and 8MB PSRAM. It supports 2.4 & 5 GHz dual-band Wi-Fi 6, BLE 5, and IEEE 802.15.4 (Zigbee, Thread) multi-protocol connectivity, meeting access requirements across a variety of network scenarios. It integrates a 2.0" color LCD display (resolution 320 × 240) with a highly sensitive capacitive touch panel, and includes an onboard buzzer, microSD card slot, and RX8130CE real-time clock. The housing is made of PC + ABS (with UV powder), providing resistance to UV aging. Internally, the device offers I2C, GPIO, UART, and RS485 expansion interfaces as well as a battery connector, which are routed externally through the M12 connectors on both sides of the body for additional expansion. The system uses M5PM1 + M5IOE1 for multi-level power management and supports low-power wake-up. The device provides standard water and dust resistance (not designed for immersion), making it suitable for a wide range of applications such as outdoor edge data collection, smart buildings, and industrial gateways.

Tutorial

Arduino IDE

This tutorial will introduce how to program and control the ToughC5 device via Arduino IDE.

Features

- ESP32-C5HR8 Core Main Controller
  - 16MB Flash

    - 8MB PSRAM
    - 2.4 & 5 GHz Dual-Band Wi-Fi 6
    - Supports BLE 5 / IEEE 802.15.4 (Zigbee, Thread)
- Human-Machine Interaction
    - 2.0" LCD Capacitive Touch Screen, 320 x 240 Resolution
    - Onboard Passive Buzzer
- Clock and Power
    - Onboard RX8130CE RTC Clock
    - RTC Clock Battery (CR2032 Coin Cell Battery DC 3V@220mAh)
    - M5PM1 + M5IOE1 Multi-Level Power Management
    - DC 3.7V Battery Interface
- Interfaces and Expansion
    - USB Type-C for Debugging / Program Download
    - Internal Interface Expansion Board: HY2.0-4P x 4 (RS485/I2C/GPIO/UART)
    - 2x M12 Body Cable Routing Holes
    - microSD Card Slot
- Easy Wiring, Magnetic Mounting
- PC + ABS (with UV Powder) Housing Material, Resistant to UV Aging

Includes

- 1 x ToughC5
- 1 x USB Type-C Cable (1m)
- 1 x Hex Key L-Shape 2.5mm (For M3 Screw)
- 1 x Hex Key L-Shape 1.5mm (For M2 Screw)
- 1 x 1.25-2P Single-Head Terminal Cable
- 1 x 1.25-4P Single-Head Terminal Cable
- 1 x Waterproof O-Ring

Applications

- Outdoor edge data collection
- Outdoor industrial field control
- Outdoor environment gateway monitoring
- Field low-power sensor terminals

Specifications

Specification	Parameter
SoC	ESP32-C5HR8 @ RISC-V® 32-bit single-core processor, up to 240MHz
Flash	16MB
PSRAM	8MB Quad
Wireless Connectivity	2.4 & 5 GHz dual-band Wi-Fi 6 / BLE 5 / IEEE 802.15.4 (Zigbee, Thread)
Input Voltage	USB (DC 5V@500mA) / RS485 interface input power (DC 6 ~ 24V@1A) / Battery (DC 3.7V)
Interface	USB Type-C x 1, HY2.0-4P x 4 (RS485/I2C/GPIO/UART)
RS485 Signal Conversion	RS485 @ SP485EEN-L/TR
LCD Screen	2.0" @ 320 x 240 ILI9342C IPS panel, maximum brightness 853nit
Touch Panel Driver IC	CHSC6540
RTC	RTC Chip: RX8130CE RTC Battery Spec: DC 3V@220mAh, Size 20 x 3.2mm Battery Socket Spec: SH1.0-2P
Battery Connector Spec	MX1.25-2P
Buzzer	Onboard passive buzzer @8530
FPC Antenna	2.4 GHz & 5 GHz, IPEX-2nd Gen, 14.3x5.3mm, cable length 50mm
Power Management	M5PM1 + M5IOE1
Power Consumption	Shutdown state: DC 4.2V@10.41uA Full load state: DC 4.2V@320mA
Housing Material	PC + ABS (with UV powder)
Product Size	75.9 x 58.0 x 42.5mm
Product Weight	119.0g
Package Size	101.0 x 83.0 x 46.0mm
Gross Weight	161.4g

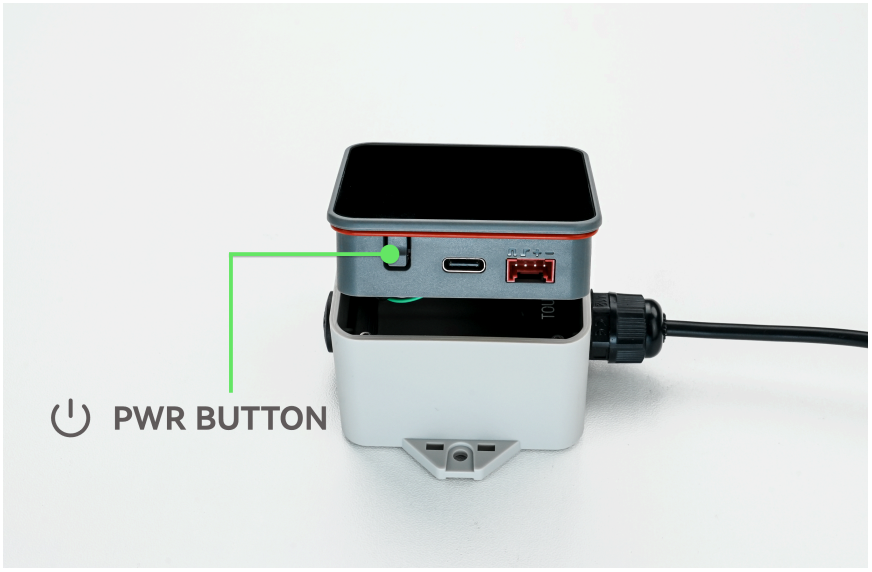
Learn

Button Operation

Enter Download Mode

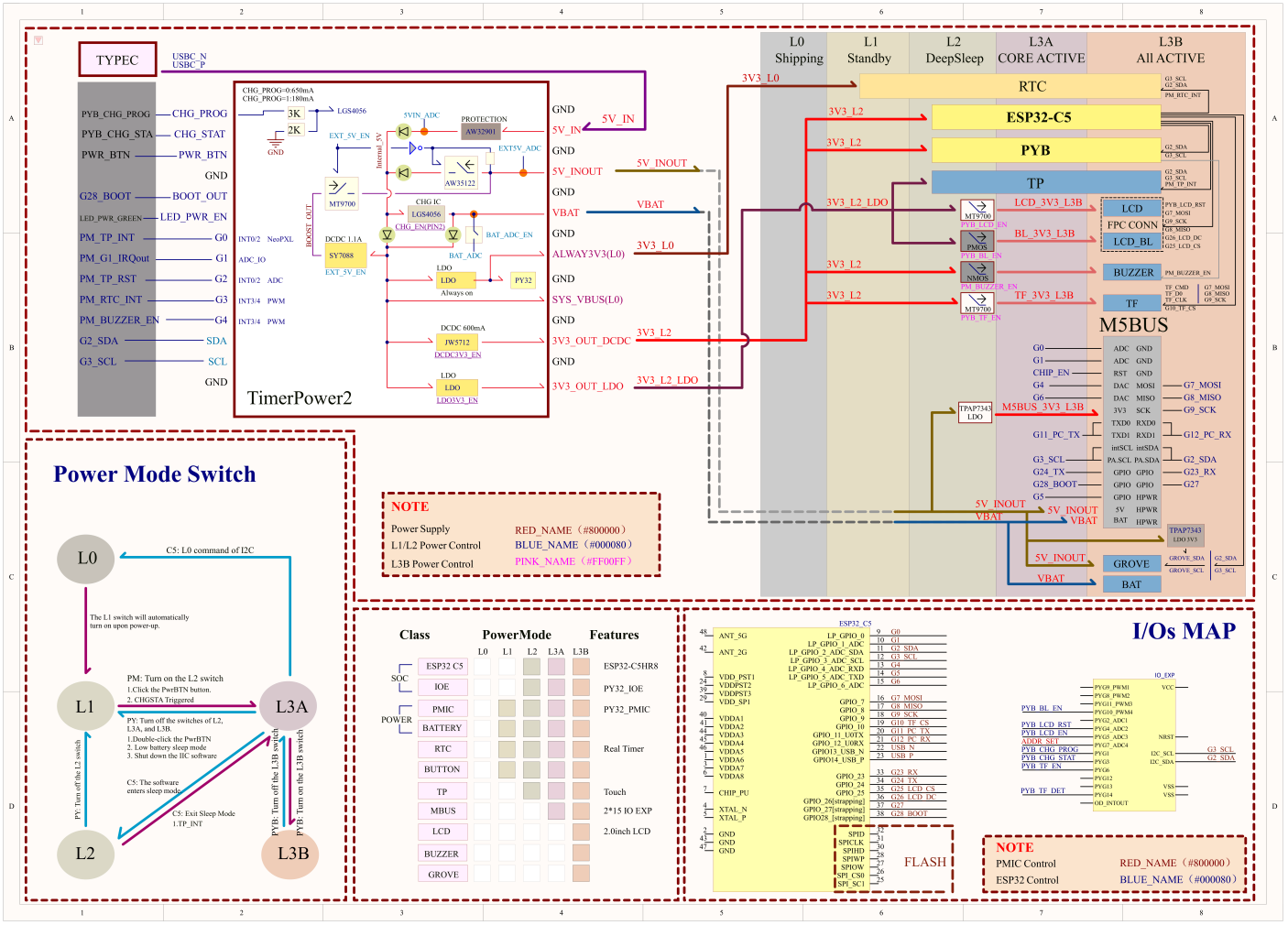
Under normal circumstances, the device can be flashed with programs directly via the USB cable. During development and debugging, if an abnormal program flash occurs, the housing can be removed and the **power button** long-pressed to force entry into download mode.

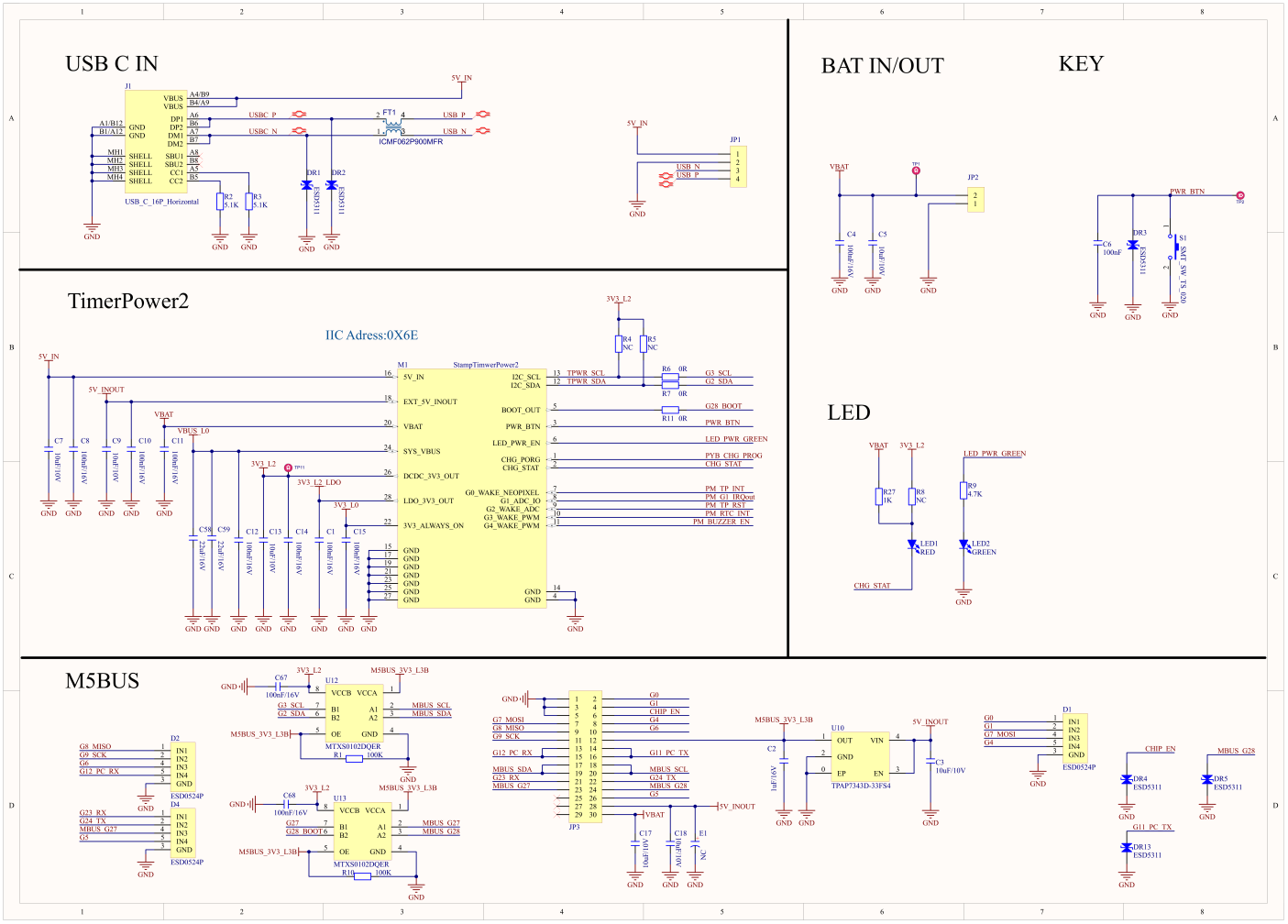
- Single click: Power on / Reset
- Double click: Power off
- Long press: Enter download mode

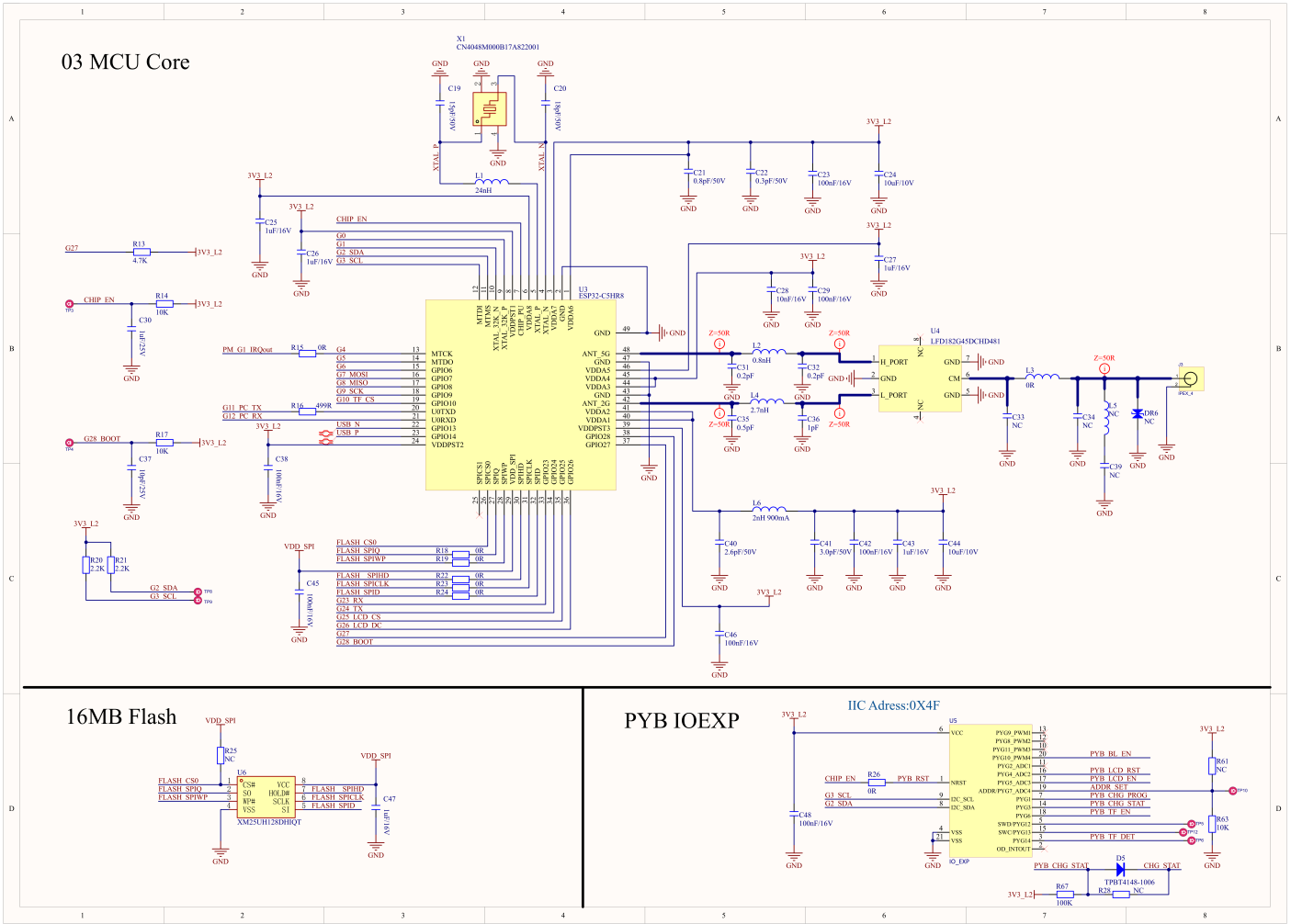


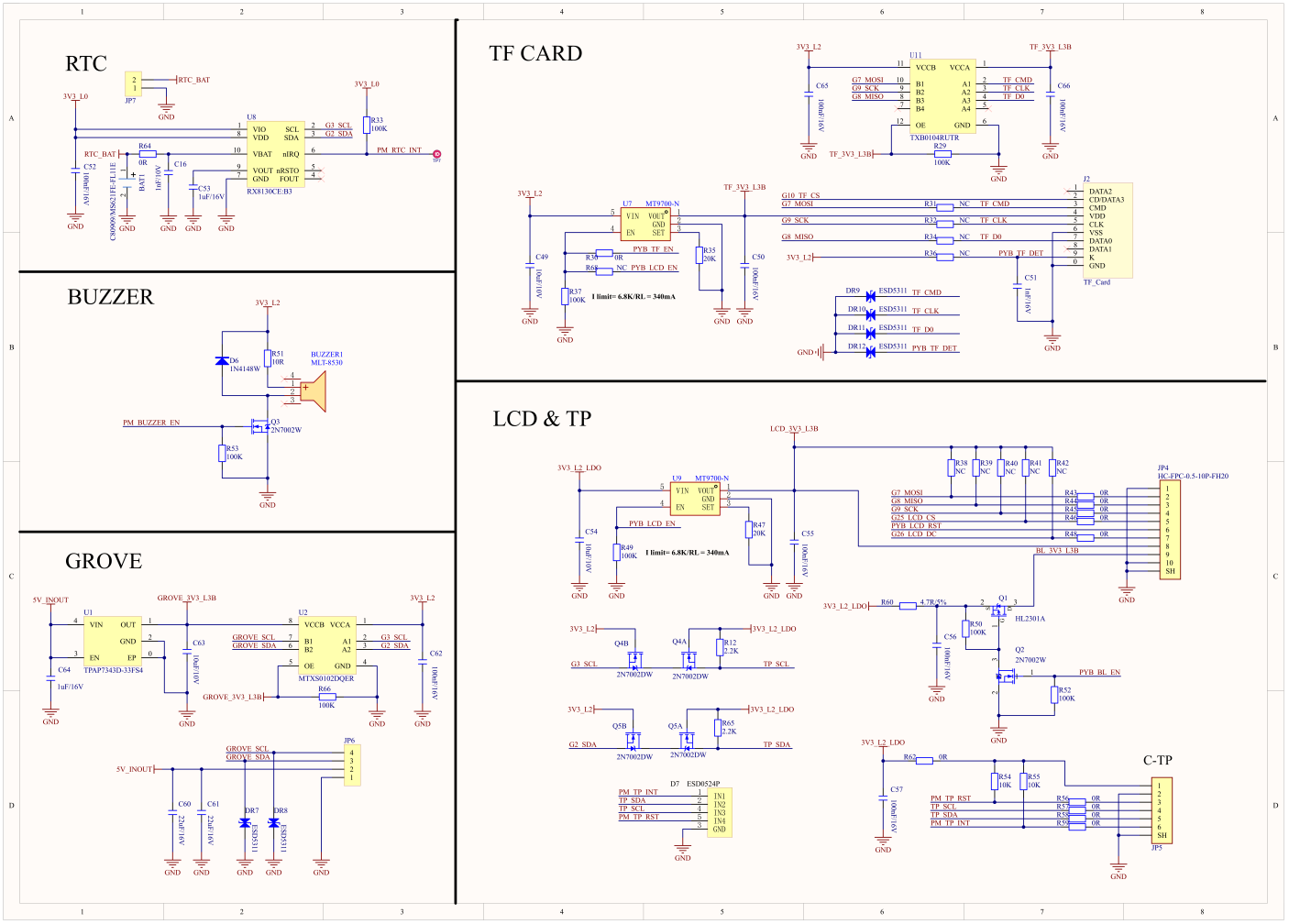
Schematics

- ToughC5 Schematics PDF




USB C IN
BAT IN/OUT
KEY
TimerPower2
IIC Adress:0X6E
LED
M5BUS


03 MCU Core
16MB Flash
PYB IOEXP
IIC Adress:0X4F



PinMap

LCD

ESP32-C5HR8	G25	G26	G7	G8	G9
ILI9342C	CS	DC	MOSI	MISO	SCK

The screen enable is controlled via `PYG5 (PYB_LCD_EN)` of the `M5IOE1` expansion chip, and the screen reset is controlled via `PYG4 (PYB_LCD_RST)`.

microSD

ESP32-C5HR8	G7	G8	G9	G10
microSD	MOSI	MISO	SCK	CS

The microSD card slot enable is controlled via `PYG6 (PYB_TF_EN)` of the `M5IOE1` expansion chip, and `PYG14 (PYB_TF_DET)` detects the microSD card insertion status.

PYG14_PYB_TF_DET	Status
LOW	Inserted
HIGH	Not Inserted

Touch

ESP32-C5HR8	G2	G3
CHSC6540	SDA	SCL

The touch screen interrupt signal input is captured via `PMG0 (PM_TP_INT)` of the `M5PM1` power management chip, and `PYG2 (PM_TP_RST)` controls the touch driver chip reset.

I2C Devices (TP & M5IOE1 & M5PM1)

ESP32-C5HR8	G2	G3
CHSC6540 (0x2E)	SDA	SCL
M5IOE1 (0x4F)	SDA	SCL
M5PM1 (0x6E)	SDA	SCL

RS485

ESP32-C5HR8	G23	G24
SP485EEN-L/TR	RS485_RX	RS485_TX

HY2.0-4P

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	G2	G3
PORT.B	GND	5V	G6	G1
PORT.C	GND	5V	G12	G11
PORT.RS485	GND	DC 6 ~ 24V INPUT	A	B

M5-Bus

FUNC	PIN	LEFT	RIGHT	PIN	FUNC
	GND	1	2	G0	ADC
	GND	3	4	G1	ADC
	GND	5	6	RST	EN
MOSI	G7	7	8	G4	DAC
MISO	G8	9	10	G6	DAC
SCK	G9	11	12	3V3	
RXD0	G12_PC_RX	13	14	G11_PC_TX	TXD0
RXD2	G12_PC_RX	15	16	G11_PC_TX	TXD2
Int SDA	G2	17	18	G3	Int SCL
PORT.A SDA	G2	19	20	G3	PORT.A SCL
GPIO	G23	21	22	G24	GPIO
GPIO	G27	23	24	G28	GPIO
	NC	25	26	G5	GPIO
	NC	27	28	5V	
	NC	29	30	BAT	

M5PM1

M5PM1	PYG0	PYG1	PYG2	PYG3	PYG4
TOUCH	PM_TP_INT		PM_TP_RST		
ESP32-C5HR8		PM_G1_IRQout			
RTC				PM_RTC_INT	
BUZZER					PM_BUZZER_EN

- PYG0(PM_TP_INT): Touch screen interrupt wake-up signal
- PYG1(PM_G1_IRQout): Interrupt output signal connected to ESP32-C5
- PYG2(PM_TP_RST): Touch screen reset
- PYG3(PM_RTC_INT): RTC interrupt wake-up signal
- PYG4(PM_BUZZER_EN): Buzzer drive signal

M5IOE1

M5IOE1	PYG1	PYG3	PYG4	PYG5	PYG6	PYG10	PYG14
CHARGE	PYB_CHG_PROG	PYB_CHG_STAT					
LCD			PYB_LCD_RST	PYB_LCD_EN		PYB_BL_EN	
microSD					PYB_TF_EN		PYB_TF_DET

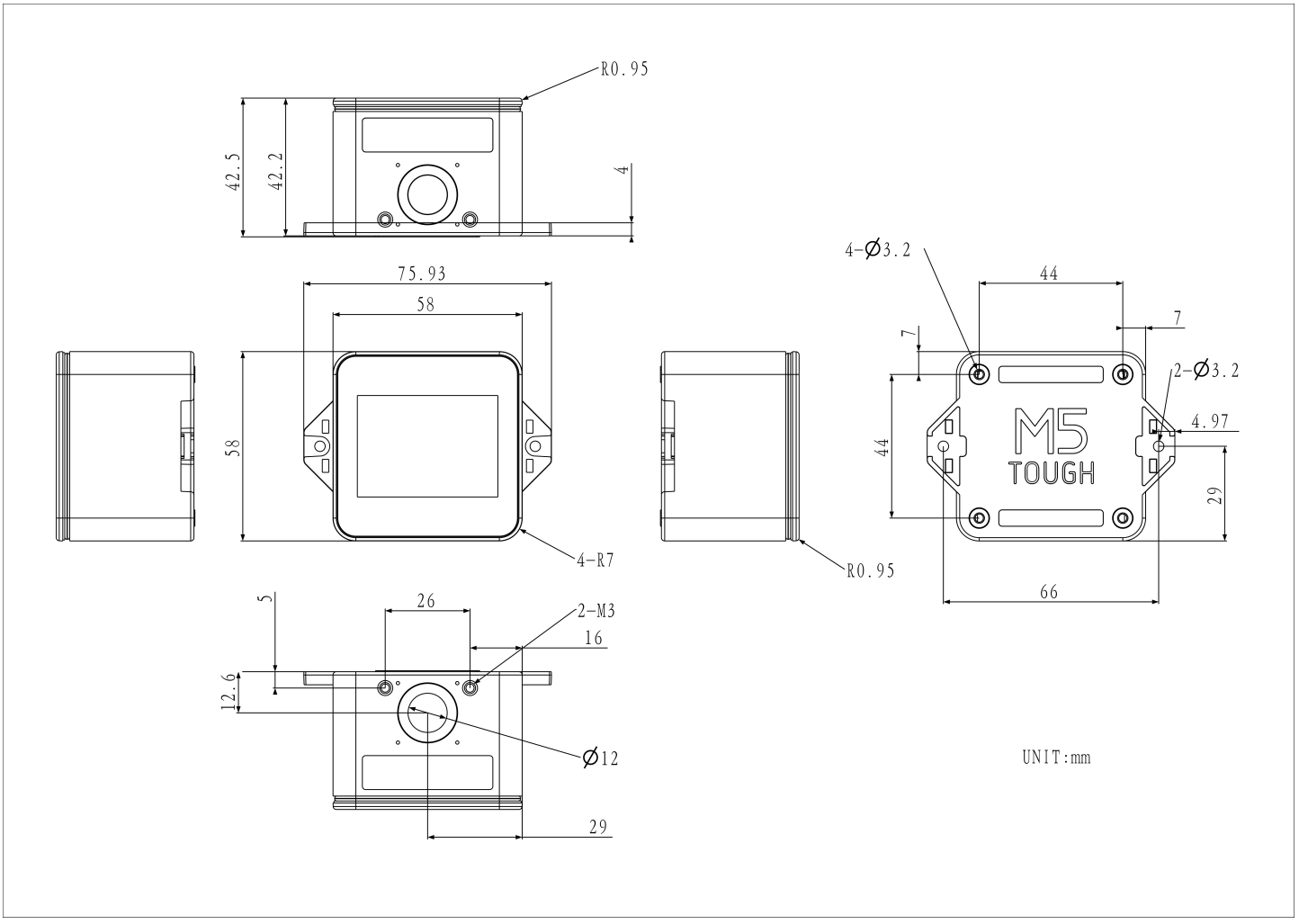
- PYG1(PYB_CHG_PROG): Charge current control

- PYG3(PYB_CHG_STAT): Charge status detection
- PYG4(PYB_LCD_RST): LCD screen reset
- PYG5(PYB_LCD_EN): LCD screen enable
- PYG10(PYB_BL_EN): LCD screen backlight drive signal
- PYG6(PYB_TF_EN): microSD card slot enable
- PYG14(PYB_TF_DET): microSD card slot insertion status detection

PYG1_CHG_PROG	CHARGE CURRENT
LOW	650mA
HIGH	180mA (Default)

Model Size

- ToughC5 Model Size PDF



Datasheets

- ESP32-C5
- ILI9342C
- RX8130CE Datasheet
- RX8130CE Register Datasheet

Softwares

Arduino

- ToughC5 Arduino Quick Start
- ToughC5 Arduino M5Unified Driver Library
- ToughC5 Arduino M5GFX Driver Library
- ToughC5 M5PM1 & M5IOE1 Power Management
- M5PM1 Arduino Driver Library
- M5IOE1 Arduino Driver Library

PlatformIO

```
[env:m5stack-toughc5]
platform = https://github.com/pioarduino/platform-espressif32.git#55.03.37
upload_speed = 1500000
monitor_speed = 115200
build_type = debug
framework = arduino
board = esp32-c5-devkitc-1
build_flags =
    -DBOARD_HAS_PSRAM
    -DCORE_DEBUG_LEVEL=5
    -DARDUINO_USB_CDC_ON_BOOT=1
    -DARDUINO_USB_MODE=1
lib_deps =
    M5Unified = https://github.com/M5Stack/M5Unified
    M5GFX = https://github.com/M5Stack/M5GFX
```

- ToughC5 Factory Firmware Source Code

Other

- ToughC5 Internal Firmware WebBurner

Video

Product Comparison

Product Comparison		
	ToughC5	**Tough**
SoC	ESP32-C5HR8 @ RISC-V 32-bit single-core processor, up to 240MHz	ESP32-D0WDQ6-V3 @ Xtensa 32-bit LX6 dual-core processor, up to 240MHz
SRAM	384 KB HP SRAM + 16 KB LP SRAM	520KB
Flash	16MB	16MB
PSRAM	8MB Quad	8MB Quad
Wireless Connectivity	2.4 & 5 GHz dual-band Wi-Fi 6 / Zigbee / Thread (802.15.4)	2.4 GHz Wi-Fi
Antenna	FPC antenna, 2.4G & 5G, IPEX-2nd Gen, 14.3 x 5.3mm, cable length 50mm	3D Wi-Fi antenna
LCD Screen	2.0" 320 x 240 ILI9342C full-color display, maximum brightness 853nit	2.0" 320 x 240 ILI9342C full-color display, maximum brightness 853nit
Touch Panel Driver IC	CHSC6540	CHSC6540
Audio	Onboard passive buzzer @8530	NS4168 16bit I2S amplifier + 1W speaker
PMU	M5PM1	AXP192
IO Expansion	M5IOE1 peripheral control system	/
RTC Clock Chip	RX8130CE	BM8563